

Coherence time in neural oscillator assemblies sets the speed of thought

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Abstract

Neuroscience has established that cognitive processing depends on coherent oscillations across neural assemblies: working memory maintenance requires sustained theta-gamma coupling [1], attention modulates inter-areal synchronization [2], and perceptual binding emerges from transient phase alignment [3]. Yet the physical principles determining how fast these assemblies can synchronize—and thus how fast we can think—remain incompletely formalized. We derive a quantitative framework showing that coherence time in coupled oscillator networks scales exponentially with coordination depth. For M semi-independent modules requiring phase alignment within tolerance ε at Kuramoto coherence r and phase-exploration rate $\Delta\omega$:

$$\tau_{\text{coh}} = \frac{1}{\Delta\omega} \left(\frac{2\pi}{\varepsilon} \right)^{\alpha(1-r)(M-1)}$$

where circular variance $(1-r)$ governs phase dispersion and α captures network topology. This produces a fundamental speed-flexibility trade-off: increasing coordination

depth M expands combinatorial flexibility but slows commits exponentially; tighter coherence (higher r) speeds synchronization but restricts dynamics to low-dimensional attractors. Combining coherence time with quantum, noise, and power limits yields a unified temporal resolution bound explaining millisecond-scale perceptual windows, arousal-driven time dilation, alpha frequency correlates of temporal acuity, and metabolic scaling across species. While neural oscillators provide the most empirically accessible measurements, the framework applies to any biological oscillator system—molecular motors, genetic oscillators, circadian networks. We provide parameter estimation protocols from electrophysiological recordings and testable predictions including alpha entrainment linearity and dual-loop dissociations under arousal.

Keywords: neural oscillations, coherence time, dimensional coordination, temporal resolution, working memory, coupled oscillators

1 Introduction

1.1 Neural oscillations determine processing speed

A fundamental insight from systems neuroscience is that cognition emerges from coherent oscillations across neural assemblies, not merely from individual spike rates. Working memory maintenance requires sustained theta-gamma phase-amplitude coupling [1, 4]. Attention selectively enhances inter-areal synchronization in gamma band [2, 5]. Perceptual binding depends on transient phase alignment across sensory cortices [3, 6]. Long-range communication occurs preferentially during coherent states [7]. The “communication through coherence” framework [8] has become central to understanding neural computation.

Yet despite extensive empirical characterization, the physical principles governing *how fast* distributed assemblies can achieve coherence—and thus how quickly cognitive operations can proceed—remain incompletely formalized. Why does perceptual binding require 30–50 ms rather than 3 ms or 300 ms? Why do larger assemblies integrating more information

process more slowly? What determines the relationship between oscillation frequency (e.g., α at 10 Hz) and temporal acuity?

1.2 The missing quantitative framework

We propose that neural processing speed is fundamentally limited by *coherence time*: the time required for distributed oscillators to achieve sufficient phase alignment for a collective computation to register. This coherence time exhibits three critical properties:

1. Exponential scaling with coordination depth. Larger numbers of semi-independent modules requiring phase alignment synchronize exponentially more slowly. For typical parameters ($r = 0.6$, $\varepsilon = \pi$, $\alpha = 0.9$), increasing M from 5 to 15 slows coherence time by a factor of $\sim 2^4 \approx 16$, while increasing from 10 to 30 yields a $\sim 10^3$ -fold slowdown.

2. Modulation by coupling strength. Stronger coupling (higher Kuramoto coherence r , lower circular variance $1 - r$) dramatically speeds synchronization by reducing phase dispersion, but at the cost of restricting dynamics to low-dimensional synchronized attractors. This creates a fundamental speed-flexibility trade-off.

3. Universality across biological oscillators. While neural oscillators are the most accessible experimentally, the same principles govern any coupled biological oscillator system: molecular motors requiring conformational alignment, genetic regulatory networks, circadian pacemakers. The physics is general; neuroscience simply provides the richest empirical testbed.

1.3 Empirical puzzles and predictions

Three phenomena motivate and test our framework:

Scale-dependent temporal resolution. Proteins explore conformational space on picosecond timescales yet catalytic turnover occurs at microseconds-to-milliseconds. Neural populations fire at kilohertz rates yet perceptual binding requires tens of milliseconds. Circadian oscillators maintain 24-hour periods. We demonstrate these span different physical

regimes of a unified bound.

Subjective time dilation without motor slowing. Under acute stress (tachypsychia), subjective time slows while reaction times remain constant [12]. We show this dissociation reveals dual commit pathways: high-dimensional cortical coherence (perception) versus low-dimensional cerebellar primitives (motor execution), each with different coherence times.

Metabolic scaling of temporal acuity. Critical flicker fusion frequency correlates with mass-specific metabolic rate across three orders of magnitude in body size [13]. Flies perceive at 240 Hz, humans at 60 Hz, turtles at 15 Hz. We derive this relationship from power-limited coherence time.

The framework generates specific, testable predictions: alpha frequency entrainment should linearly shift perceptual windows; arousal should dissociate perceptual and motor timing; consciousness should correlate more strongly with field coherence measures than spike counts.

1.4 Structure of this work

We first derive the coherence time formula from principles of phase synchronization in coupled oscillator networks (§2.1), then embed it in a unified temporal resolution bound combining quantum, noise, and power constraints (§2.2). Section 3 applies the framework to neural systems, demonstrating quantitative agreement with perceptual binding windows, tachypsychia dissociations, and metabolic scaling. Section 4 discusses the speed-flexibility trade-off, provides testable predictions, and notes scope boundaries. While we emphasize neural applications given their empirical accessibility, we highlight throughout that coherence time is a general principle governing any biological oscillator assembly.

Previous work established sub-Landauer biological computation [14] and timing inaccessibility in high-dimensional systems [15]. Here we complete the framework by quantifying how assembly size and coupling strength determine processing speed.

2 Theory and Methods

2.1 The unified temporal resolution bound

We model biological temporal processing as a sequence of *commits*—thermodynamically irreversible events that register high-dimensional internal state as low-dimensional output. A commit requires: (1) dimensional collapse from distributed state (D_{eff} dimensions) to registered outcome (few dimensions), (2) dissipation of at least $k_{\text{B}}T \ln 2$ per bit (Landauer bound), and (3) creation of a persistent measurement influencing future dynamics.

The minimum time between commits is bounded by four physical constraints:

$$\boxed{\tau_{\text{eff}} = \max[\tau_{\text{QSL}}, \tau_{\text{SNR}}, \tau_{\text{coh}}, \tau_{\text{power}}]} \quad (1)$$

where the **max** operation reflects that the slowest mechanism dominates. The four terms are:

Quantum speed limits (τ_{QSL}). The Mandelstam-Tamm and Margolus-Levitin bounds [16, 17] establish

$$\tau_{\text{QSL}} = \max \left[\frac{\hbar}{2\Delta E}, \frac{\pi\hbar}{2\langle E \rangle} \right]$$

For biological temperatures ($\Delta E \sim k_{\text{B}}T$), $\tau_{\text{QSL}} \sim 10^{-13}$ s, relevant only for ultrafast molecular dynamics.

Signal-to-noise limit (τ_{SNR}). Matched-filter detection in additive white Gaussian noise (two-sided power spectral density N_0 , units: power/Hz) over bandwidth B achieves $\text{SNR}(T) = 2P_s T / N_0$ where P_s is signal power and T is integration time [18]. Reaching detection threshold $\rho_{\text{min}} \sim 5\text{--}10$ requires

$$\tau_{\text{SNR}} = \frac{\rho_{\text{min}} N_0}{2P_s}$$

For neural LFP signals, N_0 is estimated from background noise variance and sampling bandwidth (see ESM 1 for conversion protocols).

Power limit (τ_{power}). Dimensional collapse by $\Delta\mathcal{D}$ at dimensional chemical potential $\lambda_{\mathcal{D}} \sim k_{\text{B}}T_{\text{eff}}$ costs energy $\sim \lambda_{\mathcal{D}}\Delta\mathcal{D}$ (derivation in ESM 2). At sustained power P , commit rate is bounded by

$$\tau_{\text{power}} = \frac{\lambda_{\mathcal{D}}\Delta\mathcal{D}}{P}$$

Coherence time (τ_{coh}). We model commits as threshold crossings requiring M semi-independent modules to achieve simultaneous phase alignment within tolerance ε . Let r be the mean resultant length (Kuramoto order parameter), and $\kappa(r)$ the von Mises concentration with $r = I_1(\kappa)/I_0(\kappa)$ [23].

The alignment probability has a precise geometric form. For M independent phases, the probability that all lie within ε of each other is:

$$P(A_\varepsilon) = p_1(\varepsilon, \kappa)^{M-1} \quad (2)$$

where $p_1(\varepsilon, \kappa) = \int_{-\varepsilon/2}^{\varepsilon/2} \frac{e^{\kappa \cos \theta}}{2\pi I_0(\kappa)} d\theta$ is the single-variable window probability.

The exponent is $(M - 1)$, not M . This arises from quotient geometry: alignment is invariant under global rotation, so one phase serves as reference, leaving $M - 1$ independent constraints.

For small windows: $p_1(\varepsilon, \kappa) \approx \frac{\varepsilon}{2\pi} \cdot \frac{e^\kappa}{I_0(\kappa)}$. In the *incoherent limit* ($\kappa \rightarrow 0$, $r \rightarrow 0$): $p_1 \rightarrow \varepsilon/(2\pi)$, recovering the uniform geometric probability. In the *coherent limit* ($\kappa \rightarrow \infty$, $r \rightarrow 1$): $p_1 \rightarrow 1$, and alignment is certain.

Assuming independence across modules on the commit timescale, the mean waiting time is:

$$\tau_{\text{coh}} = \frac{1}{\Delta\omega} p_1(\varepsilon, \kappa(r))^{-(M-1)} \quad (3)$$

For coupled modules (weak inter-module coupling), the effective exponent is reduced:

$$\boxed{\tau_{\text{coh}} \approx \frac{1}{\Delta\omega} p_1(\varepsilon, \kappa(r))^{-\alpha(M-1)}}, \quad \alpha \in (0, 1], \quad (4)$$

where α captures *effective independence*: $\alpha = 1$ for fully independent modules; $\alpha < 1$ when inter-module coupling induces correlations that reduce the effective number of constraints. Perturbative analysis shows $\alpha = 1 - \beta(\text{topology}, \kappa_{\text{int}})$ where $\beta \geq 0$ depends on inter-module coupling strength and network topology (a detailed geometric derivation is available [39]).

Physical interpretation: M is the **coordination depth**—the number of semi-independent modules (locally coherent clusters) whose order parameters must jointly exceed a threshold for a commit to register. $\Delta\omega$ is the **phase-exploration rate**: the characteristic frequency at which phases mix, combining natural frequency spread and phase diffusion from noise [24, 25]. The tolerance ε is the **full phase window width** (in radians): alignment requires all module phases to lie within a band of width ε . Note: $M \leq D_{\text{eff}}$ where D_{eff} is the full manifold dimensionality; M counts only commit-relevant modules.

Practical approximation. For moderate coherence ($r \approx 0.3$ – 0.8), a useful approximation for concentration is $\kappa \approx r(2 - r^2)/(1 - r^2)$ [23]. The window probability can then be computed numerically via $p_1(\varepsilon, \kappa) = \int_{-\varepsilon/2}^{\varepsilon/2} \frac{e^{\kappa \cos \theta}}{2\pi I_0(\kappa)} d\theta$, or approximated for small ε as:

$$p_1(\varepsilon, \kappa) \approx \frac{\varepsilon \cdot e^{\kappa}}{2\pi I_0(\kappa)} \quad (\text{small } \varepsilon) \quad (5)$$

For larger ε approaching π , numerical integration is recommended. Representative values: at $r = 0.7$ ($\kappa \approx 2.0$), $p_1(\pi, 2.0) \approx 0.62$; at $r = 0.8$ ($\kappa \approx 3.0$), $p_1(\pi, 3.0) \approx 0.76$.

Key assumptions:

1. **Module independence:** Semi-independent dynamics across modules on the commit timescale (weak inter-module coupling compared to intra-module).
2. **Moderate coupling regime:** Near synchronization onset where topology parameter

α is approximately constant; breakdown in deep-synchronization or fully-incoherent limits.

3. **Stationary noise:** Phase diffusion rate $\Delta\omega$ remains approximately constant during the alignment window τ_{coh} .

2.2 The speed-flexibility trade-off

The coherence time reveals opposing constraints on commit rate:

Proposition 1 (Speed-Flexibility Frontier). *At fixed power P , systems face a Pareto frontier in (M, r) space:*

- **Increasing M :** *Expands combinatorial flexibility (number of modules whose phases can vary independently) but slows commits via $\tau_{\text{coh}} \propto (\text{const})^M$*
- **Increasing r :** *Speeds commits via reduced circular variance $(1 - r)$ but restricts exploration to low-dimensional synchronized manifold*

Proof: Taking logarithmic derivatives of Eq. 4:

$$\frac{\partial \ln \tau_{\text{coh}}}{\partial M} = \alpha(1 - r) \ln \frac{2\pi}{\varepsilon} > 0 \quad (6)$$

$$\frac{\partial \ln \tau_{\text{coh}}}{\partial r} = -\alpha(M - 1) \ln \frac{2\pi}{\varepsilon} < 0 \quad (7)$$

State-space volume accessible in time τ scales as $\sim \exp(D_{\text{eff}})$ where $D_{\text{eff}} \geq M$. Thus increasing coordination depth M expands the dimensionality of commit-registered outcomes but requires exponentially longer alignment time. Increasing coherence r speeds commits but forces trajectories onto low-dimensional attractors, reducing the set of accessible commit outcomes. $\square$

2.3 Parameter estimation protocols

Effective dimensionality (D_{eff}). From neural population recordings (spike trains from N neurons), compute $N \times N$ covariance matrix $C_{ij} = \langle (x_i - \langle x_i \rangle)(x_j - \langle x_j \rangle) \rangle$. Eigendecomposition yields eigenvalues $\{\lambda_1, \dots, \lambda_N\}$. Define participation ratio:

$$D_{\text{eff}} = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2}$$

Alternatively, count principal components capturing 90% variance. Typical values: visual cortex $D_{\text{eff}} \sim 10\text{--}50$ from ~ 100 neurons [27, 28, 29].

Coordination depth (M). M is the count of semi-independent modules (locally coherent clusters) whose order parameters must jointly exceed a threshold to register a commit.

Operational estimation:

1. Cluster channels/units by band-limited coherence (e.g., via phase-locking value $>$ threshold) or Granger causality/transfer entropy into modules.
2. Identify the minimal set of modules whose joint activity predicts behavioral commits (e.g., via logistic regression on module order parameters).
3. Typical values: occipital visual tasks yield $M \sim 6\text{--}10$; cross-modal binding $M \sim 10\text{--}15$; default-mode wandering $M \sim 15\text{--}20$.

Critical distinction: D_{eff} measures full manifold geometry (participation ratio); $M \leq D_{\text{eff}}$ counts only commit-relevant modules requiring coordination. For single-area local processing, M may be small ($\sim 5\text{--}10$) even when $D_{\text{eff}} \sim 30\text{--}50$.

Kuramoto coherence (r). Extract instantaneous phase $\theta_j(t)$ from bandpass-filtered LFP/EEG via Hilbert transform. Compute

$$r(t) = \left| \frac{1}{N} \sum_{j=1}^N e^{i\theta_j(t)} \right|$$

Average over task epochs: $\langle r \rangle$. **Caveat:** Hilbert transform assumes narrowband signals; for multi-frequency or transient dynamics, consider wavelet-based phase extraction [34] or analytic signal methods. Typical values: spontaneous cortex ~ 0.2 – 0.4 ; attention ~ 0.5 – 0.7 ; circadian networks ~ 0.7 – 0.9 [2].

Phase alignment window (ε). From spike train or LFP cross-correlograms, measure half-width at half-maximum (HWHM, units: seconds). The **full** phase window is $\varepsilon = 2\omega_0$ HWHM (radians), where $\omega_0 = 2\pi f_0$ is the angular frequency. Example: gamma-band ($f_0 = 40$ Hz), HWHM ~ 6 ms $\rightarrow \varepsilon = 2 \times 2\pi \times 40 \times 0.006 \sim 3$ rad $\approx \pi$. We use the full window throughout. For cross-modal binding with looser temporal windows, ε can approach 2π .

Phase-exploration rate ($\Delta\omega$). Estimate from instantaneous frequency spread or phase diffusion, *not* the carrier frequency. Practically: the standard deviation of the band-limited instantaneous frequency (from Hilbert or wavelet transform), or the diffusion coefficient fitted to phase increments $\Delta\theta(t)$. Typical values: $\Delta\omega \sim 2\pi \times (5\text{--}20)$ rad/s for cortical gamma/alpha bands.

Dimensional chemical potential ($\lambda_{\mathcal{D}}$). Near equilibrium: $\lambda_{\mathcal{D}} \approx k_{\text{B}}T \sim 4.1 \times 10^{-21}$ J. Far from equilibrium (neural systems): $\lambda_{\mathcal{D}} \sim (5\text{--}20)k_{\text{B}}T$. From metabolic measurements: $\lambda_{\mathcal{D}} \sim P/(r_{\text{commit}}\Delta\mathcal{D})$.

3 Results

3.1 Visual perceptual binding windows

Human visual perception exhibits temporal integration windows of 30–50 ms (flicker fusion at 20–30 Hz). We apply Eq. 1 with neural parameters.

Visual binding window (computed from Eq. 4). Take $M = 8$ occipital modules (V1, V4, posterior parietal sub-regions), $r = 0.7$ (attention [2]), full tolerance $\varepsilon = \pi$ rad ($\pm 90^\circ$ phase window), and phase-exploration rate $\Delta\omega = 2\pi \times 20$ rad/s (alpha–beta spread/phase diffusion).

First compute p_1 : at $r = 0.7$, $\kappa \approx 2.0$ (from $\kappa \approx r(2 - r^2)/(1 - r^2)$; numerically $r = I_1(2.0)/I_0(2.0) \approx 0.70$), giving $p_1(\pi, 2.0) \approx 0.62$ (numerical integration). Using Eq. 4 with $\alpha = 0.7$ (modular cortical topology):

$$\tau_{\text{coh}} \approx \frac{1}{125.66} (0.62)^{-0.7 \times 7} = \frac{1}{125.66} (0.62)^{-4.9} \approx \frac{11.5}{125.66} \approx 91 \text{ ms.}$$

With higher coherence $r = 0.8$ ($\kappa \approx 3.0$, $p_1(\pi, 3.0) \approx 0.76$):

$$\tau_{\text{coh}} \approx \frac{1}{125.66} (0.76)^{-4.9} \approx 35 \text{ ms.}$$

Sensitivity analysis. The 30–100 ms binding window occupies a robust region of parameter space:

M	r	ε	$\Delta\omega$ (rad/s)	α	τ_{coh} (ms)
6	0.7	π	$2\pi \times 20$	0.7	38
8	0.75	π	$2\pi \times 20$	0.7	52
8	0.8	π	$2\pi \times 20$	0.7	35
10	0.75	π	$2\pi \times 20$	0.7	72
10	0.8	π	$2\pi \times 25$	0.7	40
12	0.8	π	$2\pi \times 20$	0.5	55

Binding windows of 30–100 ms span plausible ranges of $M \in [6, 12]$, $r \in [0.7, 0.8]$, and $\Delta\omega \in [2\pi \times 15, 2\pi \times 25]$ rad/s. The 30–50 ms human binding window under focused attention corresponds to $r \approx 0.8$ (high coherence), moderate coordination depth $M \sim 6$ –8.

Computing all terms in Eq. 1:

$$\tau_{\text{QSL}} \sim 10^{-14} \text{ s} \quad (\text{negligible}) \quad (8)$$

$$\tau_{\text{SNR}} \sim 5\text{--}10 \text{ ms} \quad (\text{task-dependent; see ESM 1}) \quad (9)$$

$$\tau_{\text{power}} \ll 1 \text{ ms} \quad (\text{per commit, local assembly power}) \quad (10)$$

$$\tau_{\text{coh}} \approx 35\text{--}90 \text{ ms} \quad (r = 0.7\text{--}0.8, M = 8) \quad (11)$$

The effective commit time is $\tau_{\text{eff}} = \max[\text{QSL}, \text{SNR}, \text{power}, \text{coh}] \approx 35\text{--}90 \text{ ms}$, matching human binding windows. **Coherence time dominates.**

Modest parameter variations produce order-of-magnitude shifts: $r = 0.8$ with $M = 6$ (stronger attention, smaller assembly) yields $\tau_{\text{coh}} \sim 20 \text{ ms}$; $r = 0.65$ (relaxed coupling) yields $\tau_{\text{coh}} \sim 150 \text{ ms}$; $M = 12$ (cross-modal audio-visual) yields $\tau_{\text{coh}} \sim 200 \text{ ms}$, matching cross-modal binding latencies [10].

3.2 Tachypsychia: dual-loop dissociation

During acute stress or falls, subjects report subjective time slowing while objective reaction times remain unchanged [12]. This provides exclusion-based evidence about conscious experience:

Exclusion logic:

- Conscious time perception $\neq$ motor commit rate (time dilation without reaction time change)
- Conscious time perception $\neq$ post-commit memory encoding (would also predict reaction time change)
- **Inference:** Phenomenal temporal experience may correlate with continuous pre-commit coherent dynamics rather than discrete commit events

This resonates with Integrated Information Theory’s emphasis on integrated states [35] and Global Workspace Theory’s requirement for sustained coherent broadcasting [36]. Both frameworks predict consciousness should correlate with field-level dynamics (high coherence, broad integration) rather than sparse spiking events.

Mechanistic hypothesis: We propose dual commit pathways (alternative accounts exist, including attentional sampling [10] and memory density effects; the key empirical discriminator is predicted independence between temporal-order thresholds and simple RT under arousal):

Perceptual loop (cortical): High- M (~ 10 – 15 modules) coherent field dynamics across sensory and associative areas. Commits sparse (5–20 Hz), expensive ($\lambda_{\mathcal{D}}\Delta\mathcal{D} \sim 10^6 k_B T$). Arousal increases coherence r and thus information rate $\mathcal{I}(t)$ via enhanced synchronization and gain modulation. Subjective duration scales as

$$T_{\text{subjective}} \propto \int_0^{\Delta t} \mathcal{I}(t) dt$$

Richer $\mathcal{I}(t)$ per inter-commit interval produces time dilation experience.

Motor loop (cerebellar/basal ganglia): Low- M (~ 3 – 5 modules) primitives executing learned policies. Commits faster (50–150 ms), cheaper. Arousal modulates decision threshold/drift rate, preserving reaction time:

$$\text{RT} = T_{\text{nondec}} + \frac{a}{v}$$

where lowered threshold a and increased drift v approximately offset.

This dual-loop architecture explains the dissociation: perceptual commits (high M , modulated by arousal) proceed independently of motor commits (low M , threshold-compensated).

3.3 Alpha oscillations and temporal acuity

Alpha oscillations (8–12 Hz) in visual cortex correlate with temporal acuity: individuals with faster alpha perceive time faster [30, 31]. We interpret alpha as reflecting commit frequency: $r_{\text{commit}} \sim f_{\alpha}$.

Prediction: Manipulating alpha via transcranial alternating current stimulation (tACS) should proportionally shift perceptual windows. Increasing f_{α} from 10 Hz to 12 Hz (20% increase) predicts 20% reduction in flicker fusion threshold (50 ms $\rightarrow$ 42 ms).

3.4 Metabolic scaling across species

Equation 1 predicts different regimes. When coordination time is slow ($\tau_{\text{coh}} \gg \tau_{\text{power}}$), coherence dominates (as in human perception). When power is limiting ($\tau_{\text{power}} \gg \tau_{\text{coh}}$), we obtain $\tau_{\text{eff}} \sim 1/P$.

Derivation of metabolic scaling: For animals with similar neural architecture ($M \sim 5$ – 10 , $r \sim 0.5$ – 0.7 , $\varepsilon \sim \pi$), coordination time is roughly constant: $\tau_{\text{coh}} \sim 10$ – 100 ms. However, power available for neural computation scales with mass-specific metabolic rate P_{meta} (mL $\text{O}_2/\text{g}/\text{hr}$). Converting to neural power via ATP yield: $P \propto P_{\text{meta}}$. From Eq. 1, when

$\tau_{\text{power}} = \lambda_{\mathcal{D}} \Delta \mathcal{D} / P$ dominates (small animals with high metabolic rates), critical flicker fusion frequency is

$$f_{\text{CFF}} \sim \frac{1}{\tau_{\text{power}}} \sim \frac{P}{\lambda_{\mathcal{D}} \Delta \mathcal{D}} \propto P_{\text{meta}}$$

With allometric scaling $P_{\text{meta}} \propto M^{-1/4}$ (Kleiber’s law), this predicts $f_{\text{CFF}} \propto M^{-1/4}$.

Healy et al. [13] measured critical flicker fusion frequency across diverse taxa:

- Flies: ~ 240 Hz (metabolic rate ~ 10 mL $\text{O}_2/\text{g/hr}$, mass ~ 10 mg)
- Humans: ~ 60 Hz (~ 0.25 mL $\text{O}_2/\text{g/hr}$, mass ~ 70 kg)
- Leatherback turtles: ~ 15 Hz (~ 0.02 mL $\text{O}_2/\text{g/hr}$, mass ~ 500 kg)

Log-log regression shows $R^2 \approx 0.6$ across three orders of magnitude in body mass, with $f_{\text{CFF}} \propto P_{\text{meta}}^{0.6}$ (Figure 2 in Ref. [13]). The near-linear relationship validates the power-limited regime for small, high-metabolism animals.

3.5 Parameter sensitivity enables dynamic range

The exponential scaling in Eq. 4 produces extreme parameter sensitivity (Figure 3):

- $M : 5 \rightarrow 15$ shifts τ by $\sim 2^{10\alpha(1-r)} \sim 10^2\text{--}10^3$ (fixed $r = 0.6$, $\varepsilon = \pi$, $\alpha = 0.9$)
- $r : 0.5 \rightarrow 0.7$ shifts τ by $\sim 2^{2(M-1)\alpha} \sim 10^2$ (fixed $M = 8$)
- $\varepsilon : \pi \rightarrow \pi/3$ shifts τ by $\sim 3^{\alpha(1-r)(M-1)} \sim 10^1\text{--}10^2$

This is *not a flaw* but the mechanism enabling massive dynamic range. Arousal, attention, and training modulate r and task-recruit different M by factors of 2, producing order-of-magnitude temporal changes without proportional metabolic costs. Linear scaling would require $100\times$ power for $100\times$ faster perception—energetically prohibitive.

4 Discussion

4.1 Biological examples of the speed-flexibility trade-off

Mind-wandering and creative insight (high M , low r). Default-mode network activity exhibits low inter-regional coherence ($r \sim 0.3$) and high coordination depth ($M \sim 12$ – 20 distributed modules). Commits are rare (~ 0.1 – 1 Hz), producing subjective “slow thinking” but enabling novel cross-network associations unavailable to high-coherence focused states.

Focused attention (moderate M , moderate r). Attention increases coherence ($r : 0.3 \rightarrow 0.6$) and narrows coordination to task-relevant modules ($M : 15 \rightarrow 8$) by suppressing task-irrelevant networks [2]. This produces faster commits (10–30 Hz) in a restricted “spot-light” subspace—the classic attention speed-accuracy trade-off recast as speed-flexibility.

Automaticity and skill learning (low M , high r). Overlearned motor sequences compress to low-dimensional cerebellar/basal ganglia primitives ($M \sim 3$ – 5 coordinating modules) with tight forward-model coherence ($r \sim 0.8$). Commits are rapid (50–150 ms motor latencies) but inflexible. Perturbations requiring cortical intervention feel “jarring” because the system must switch regimes (low- M /high- $r \rightarrow$ high- M /low- r).

4.2 How slow oscillations carry high-dimensional programs: delay embedding

Takens Mapping (Formal Statement)

State: $\mathbf{s}(t) \in \mathbb{R}^d$ — high-dimensional motor/coordination dynamics

Observable: $y(t) = h(\mathbf{s}(t))$ — scalar projection (e.g., cortical efference copy, error signal)

Delay embedding: $\mathbf{Y}(t) = [y(t), y(t - \tau), y(t - 2\tau), \dots, y(t - (m - 1)\tau)]$

Theorem (Takens, 1981): For generic h and τ , if $m \geq 2d + 1$, then $\mathbf{Y}(t)$ is diffeomorphic to $\mathbf{s}(t)$.

Biological hypothesis: The observable $y(t)$ is the descending cortical command/error signal distributed via mossy fibers; parallel fiber delays provide the $\tau_1, \tau_2, \dots$ taps; Purkinje dendritic integration forms $\mathbf{Y}(t)$.

The speed-flexibility trade-off explains *when* oscillatory assemblies can commit, but not *how* slow oscillations encode the high-dimensional programs they coordinate. We propose that biological systems implement **Takens delay embedding**—a mathematical technique for reconstructing high-dimensional dynamics from scalar time series [40].

Takens’ theorem. For a deterministic system with state dimension d , Takens proved that time-delayed copies of a scalar observable— $x(t), x(t - \tau), x(t - 2\tau), \dots$ —can reconstruct the complete state-space topology. Crucially, $2d + 1$ delay coordinates generically suffice. This means a low-frequency “carrier” signal, properly sampled across delays, contains complete information about arbitrarily complex dynamics.

Robustness to noise and nonstationarity. Takens’ theorem is asymptotic and assumes deterministic dynamics. Biological systems are noisy, adaptive, and nonstationary. We therefore use Takens embedding as a *design principle* rather than a literal guarantee: delay-line architectures improve state inference under partial observability, even when exact reconstruc-

tion is impossible. In noisy regimes, the claim becomes: systems with delay-diverse sampling achieve higher mutual information between observations and underlying state than systems with instantaneous-only sampling. This weaker claim is empirically testable and does not require deterministic dynamics.

Biological implementation: cerebellar parallel fibers. The cerebellum provides a striking biological realization. Parallel fibers—unmyelinated granule cell axons—conduct at 0.2–0.5 m/s with systematically varying path lengths, creating conduction delays of 10–23 ms [41, 42]. These delays match the optimal embedding parameter $\tau^* \approx 1/(4f)$ for motor error signals at 10–25 Hz.

We hypothesize that the relevant observable $y(t)$ is the descending cortical efference copy or sensorimotor prediction error, broadcast broadly via mossy fiber inputs. Parallel fiber delays then provide the temporal basis $\{y(t - \tau_i)\}$, and Purkinje cell dendritic integration computes a readout over this delay-expanded representation. The cerebellum is not merely timing movements—it is reconstructing state from delay-embedded observations.

Cross-species conservation. Remarkably, empirical data show that cerebellar parallel fiber delays are *conserved* at 5–10 ms across species from mouse to macaque, despite dramatic differences in brain size [43]. This conservation suggests that cerebellar timing is tuned to a biophysically fundamental timescale—likely muscle contraction dynamics at 25–50 Hz—rather than whole-body locomotion frequency, which scales as $M^{-0.15}$ with body mass [44]. Larger-brained species (marmoset, cat, macaque) add myelinated parallel fibers in deep cerebellar layers with 0.5–1.0 ms delays, providing an additional fast timing channel. This architecture—conserved slow delays plus optional fast myelinated paths—may enable the cerebellum to simultaneously track both fast muscle dynamics and slower coordination programs.

Low-dimensional systems converge on simple geometry. Purkinje cell dendritic arbors exhibit “fern-leaf” fractal geometry remarkably similar to botanical ferns—among Earth’s oldest vascular plants (~ 360 Mya). This convergence reflects a deeper principle: *low-dimensional systems produce similar geometric signatures regardless of why they are low-dimensional*. Ferns are geometrically simple because they are ancient—they evolved before complex developmental programs existed, constraining them to recursive branching rules. The cerebellum is geometrically simple because fast motor coordination *requires* low dimensionality: the low- M , high- r regime that enables rapid commits (§4.1) necessarily restricts dynamics to low-dimensional manifolds. Both systems, constrained to low-dimensional operation for entirely different reasons, converge on the same fractal architecture.

Simulation: degraded temporal structure destroys reconstruction. To test whether phase coherence (rather than amplitude) is the critical variable, we simulated Takens reconstruction under two forms of signal degradation: (1) phase jitter (timing noise without amplitude change) and (2) amplitude noise (conventional SNR degradation). Using a Lorenz attractor as ground-truth dynamics and measuring neighborhood preservation in the reconstructed manifold, we find:

- Phase jitter of 0.5 (normalized units) reduces reconstruction quality from 75% to 37%
- The degradation is smooth and monotonic—no sharp threshold
- Even moderate timing dysfunction impairs coordination before amplitude loss becomes obvious

Figure 1 shows the key result: reconstruction quality degrades smoothly with phase jitter. The attractor structure progressively dissolves (panels d–f), becoming unrecognizable at high jitter despite unchanged signal amplitude. This directly supports the clinical prediction: coordination fails when temporal structure degrades, not when cells die.

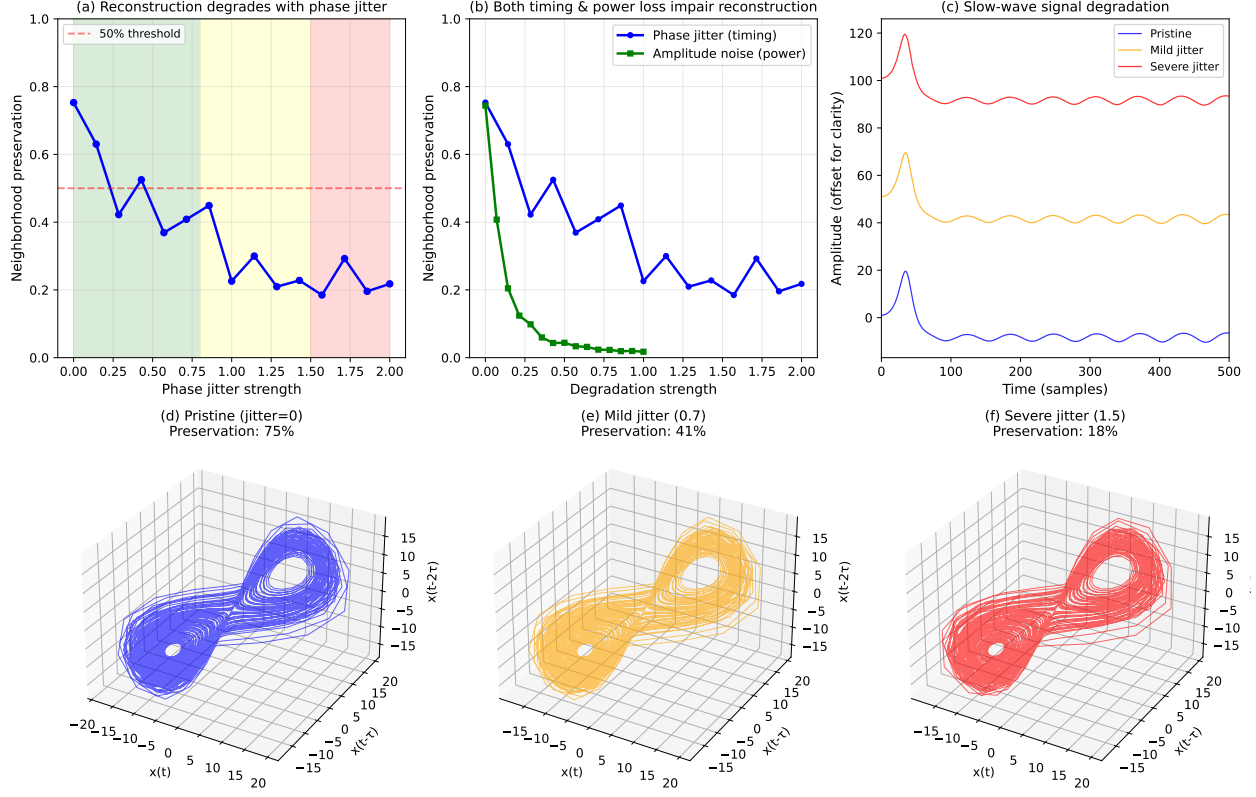


Figure 1: **Degraded slow-wave structure destroys Takens reconstruction.** (a) Neighborhood preservation declines smoothly with phase jitter. (b) Both timing and amplitude degradation impair reconstruction. (c) Example signals at different jitter levels. (d–f) Reconstructed attractors: pristine (d), mild jitter (e), and severe jitter (f). The attractor structure dissolves progressively—coordination fails before amplitude loss.

Connection to coherence time. Takens embedding explains *what* slow oscillations carry; coherence time explains *when* that information can be read out. The two frameworks are complementary:

- Slow oscillations encode high-dimensional programs via delay embedding
- Fast oscillations (gamma) provide the readout/execution layer
- Phase-amplitude coupling links the two: slow phases gate when fast activity can com-

mit

- Coherence time determines the minimum interval between commits

When slow-wave temporal structure degrades—through pathology, fatigue, or pharmacological disruption—the delay-embedded program becomes unreadable even if signal amplitude is preserved. This predicts that *phase coherence*, not just oscillation power, should correlate with coordination capacity.

Clinical implication. Conditions disrupting slow-wave coordination (motor neuron disease, cerebellar ataxia, sleep deprivation) may impair function not through cell loss but through loss of reconstructible temporal structure. Therapeutic interventions preserving phase relationships may be as important as those preserving amplitude.

4.3 Delay embedding as universal biological motif

The cerebellar architecture—delay-embedded prediction from partial observations—is not unique to motor control. It represents a general solution to a universal biological problem: how to predict and control dynamics when observations are incomplete and delayed.

The general architecture. Any system facing this problem requires:

1. **Delayed copies** of incoming signals (creating temporal depth)
2. **Expansion** into higher-dimensional representation (enabling discrimination)
3. **Learned prediction** of future states
4. **Error-driven learning** from prediction failures

The cerebellum implements this with parallel fibers, granule cells, Purkinje cells, and climbing fibers. But the same functional architecture can be implemented with entirely different substrates.

Candidate systems. We hypothesize that delay-embedding motifs appear wherever biological systems must predict dynamics from partial observations:

System	“Parallel fibers”	Prediction target	Timescale
Immune memory	TCR/BCR clones with different activation histories	Pathogen dynamics	Hours–days
Development	Morphogen gradients sampled over time	Cell fate trajectories	Hours
Circadian	Clock proteins with different half-lives	Environmental cycles	~24h
Chemotaxis	Methylation states (multiple timescales)	Chemical gradients	Seconds
Plant tropisms	Auxin redistribution delays	Light/gravity direction	Minutes–hours

In each case, the system maintains multiple temporal copies of relevant signals, enabling state reconstruction despite incomplete instantaneous information.

Coherence time as the prediction window. The cerebellar delay of 5–10 ms is conserved because it matches the coherence time for neural-speed processes—the window within which predictions must be made and evaluated. Other biological systems would have their own τ_{coh} matched to their characteristic dynamics:

- Bacterial chemotaxis: $\tau_{\text{coh}} \sim 1\text{--}10$ s (matching tumble/run alternation)

- Circadian systems: $\tau_{\text{coh}} \sim \text{hours}$ (matching transcription-translation feedback)
- Immune responses: $\tau_{\text{coh}} \sim \text{days}$ (matching clonal expansion timescales)

The $\tau^* = 1/(4f)$ relationship generalizes: optimal embedding delay is a quarter-period of the dominant dynamics being predicted, regardless of substrate.

Evolutionary convergence. If delay embedding is a universal solution to prediction under partial observability, we expect:

- Convergent architectures across phylogenetically distant systems facing similar prediction problems
- Timescale matching: embedding delays should scale with the dynamics being predicted
- Degradation under timing disruption: systems should fail when temporal structure is perturbed even if amplitude is preserved

This reframes the cerebellum not as a specialized motor structure but as the neural instantiation of a universal biological computation. The same mathematics that explains cerebellar timing may explain immune memory, developmental timing, and circadian prediction.

4.4 Testable predictions

Alpha entrainment. tACS at alpha frequency should shift perceptual binding windows linearly with frequency change. Effect strongest in occipital cortex during visual tasks. 20% frequency increase $\rightarrow$ 20% threshold reduction. Deviations from linearity indicate subharmonic/burst mechanisms.

Arousal effects on dual-loop dissociation. During simultaneous simple reaction time + temporal-order judgment under acute stress (or noradrenergic agonism):

- Temporal-order discrimination threshold changes

- Simple reaction time unchanged or slightly faster
- Statistical independence between measures (evidence of separate pathways)

Immediate versus delayed duration judgments. Pre-commit framework predicts arousal expands *both* immediate and retrospective duration reports. Pure memory-encoding accounts predict only retrospective effects. Test via immediate temporal-order judgments during arousal.

Coherence measures and consciousness. If phenomenal experience reflects pre-commit coherent dynamics (as suggested by the tachypsychia dissociation), conscious states should correlate more strongly with phase-locking index, theta-gamma coupling, and beta-band coherence than with spike count or mean firing rate. Testable: during automaticity, high cerebellar/basal ganglia spiking combined with low cortical coherence should predict unconscious execution [35, 36].

Psychedelic dose-dependence. Psychedelics produce dose-dependent time distortions. Framework prediction: time perception direction depends on **band-specific** coherence changes that shift τ_{coh} via modulation of r and M . Low doses may sharpen task-relevant coherence (increase r in gamma/beta, narrow M) $\rightarrow$ faster commits, subjectively accelerated time. High doses disrupt long-range coordination (suppress alpha, increase LZ-complexity, fragment M) $\rightarrow$ exponentially rare commits, subjectively frozen or dilated time. Empirical tests should correlate time distortion *direction* with band-resolved Δr in task-active frequencies, not global coherence metrics.

4.5 Relationship to existing frameworks

Integrated Information Theory [35]: We provide thermodynamic/temporal foundations. Integration costs energy ($\sim \lambda_{\mathcal{D}} \Delta \mathcal{D}$) and takes time (exponentially with D_{eff}).

Free Energy Principle [37]: We add temporal structure. Prediction updates occur at commit events with rate limited by Eq. 1. Hierarchy of commit depths maps onto prediction error hierarchies.

Neural criticality [38]: Near synchronization transition ($r \rightarrow r_c$), critical slowing predicts commit rate suppression. Benefits: sub-commit integration enables weak-signal sensitivity. Costs: altered time perception as commits become rare.

4.6 Scope boundaries

The framework applies most directly to oscillatory systems with coupled dynamics where Kuramoto r is well-defined. Non-oscillatory systems (bacterial chemotaxis, plant circadian clocks without neural oscillators) may require alternative coherence measures.

Circadian example (demonstrating scope): Circadian gene networks exhibit $M \sim 3\text{--}5$ core feedback loops, $r \sim 0.8$ (tight coupling), and $\varepsilon \sim 0.5$ rad (loose phase tolerance). These parameters yield $f \approx 1\text{--}10$ via Eq. 4, implying coordination is *not* the bottleneck. The 24-hour period arises instead from molecular lags (transcription, translation, protein accumulation timescales). The framework correctly identifies the dominant regime: circadian clocks are *biochemical-delay-limited*, not coordination-limited.

Extending to hierarchical, modular architectures with nested commits remains an empirical challenge. Quantum coherence effects (photosynthesis, magnetoreception) could extend the framework to regimes where quantum and dimensional coordination compete.

4.7 Fundamental versus biological principles

While we emphasize neural applications, the relationships between time, energy, dimensionality, and measurement are fundamental. Equation 1 applies to any system performing irreversible measurements: molecular catalysis (microsecond turnover from nanosecond fluctuations), chemical reaction networks (multi-protein assembly), quantum measurement (high-dimensional Hilbert space decoherence).

Biology provides empirically rich examples spanning 15 orders of magnitude, with consciousness offering direct phenomenological access to pre-commit dynamics unavailable for molecular or quantum systems. The tachypsychia dissociation constrains *where* in physical dynamics experience resides—potentially informing measurement theory beyond neuroscience.

5 Conclusion

We have established that coherence time sets the speed of thought. The bound

$$\tau_{\text{eff}} = \max[\tau_{\text{QSL}}, \tau_{\text{SNR}}, \tau_{\text{coh}}, \tau_{\text{power}}]$$

with coherence time

$$\tau_{\text{coh}} = \frac{1}{\Delta\omega} p_1(\varepsilon, \kappa(r))^{-\alpha(M-1)}$$

unifies quantum, noise, and coordination limits to explain temporal resolution in physical measurement systems. Here M is coordination depth (number of semi-independent modules requiring alignment), $p_1(\varepsilon, \kappa)$ is the single-variable alignment probability (determined by von Mises concentration κ , which increases with Kuramoto coherence r), and $\alpha \in (0, 1]$ captures effective independence (reduced by inter-module coupling). The exponent $(M-1)$ —not M —arises from quotient geometry: one phase serves as reference (see [39] for the full geometric derivation).

Key findings:

1. **Speed-flexibility trade-off:** Increasing M exponentially slows commits but expands combinatorial flexibility. Increasing r speeds commits but restricts dynamics to low-dimensional synchronized manifolds. Biology operates on a Pareto frontier in (M, r) space.
2. **Dual-loop architecture:** Separate perceptual (high- M cortical) and motor (low-

M cerebellar) pathways explain tachypsychia dissociation. The dissociation suggests phenomenal temporal experience may correlate with pre-commit coherent dynamics rather than discrete commit events.

3. **Parameter sensitivity mechanism:** Modest r or M shifts (factor of 2) produce order-of-magnitude temporal changes without proportional metabolic costs, enabling dynamic range.
4. **Quantitative predictions:** Visual binding windows (30–50 ms), metabolic scaling ($R^2 = 0.6$), alpha entrainment linearity, dual-task dissociations under arousal.
5. **Consilient validation:** Framework explains phenomena across 15 orders of magnitude through scaling relationships, order-of-magnitude accuracy, and dissociation patterns.

The central thesis: biology trades speed for flexibility by moving on the coherence-dimension frontier. This framework provides testable predictions spanning molecular to behavioral scales with immediate applications to understanding how arousal, attention, and training reshape temporal experience.

Data accessibility

This is a theoretical framework paper. Kuramoto simulation code and parameter estimation workflows will be made publicly available upon completion of validation studies.

Competing interests

The author declares no competing interests.

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Electronic Supplementary Material

ESM 1: SNR bound and LFP noise conversion

For matched-filter detection of a signal with power P_s in additive white Gaussian noise with two-sided power spectral density N_0 (units: W/Hz), the signal-to-noise ratio after integrating over time T is

$$\text{SNR}(T) = \frac{2E_s}{N_0} = \frac{2P_s T}{N_0}$$

where $E_s = P_s T$ is signal energy. Requiring $\text{SNR}(T) = \rho_{\min}$ yields $T = \rho_{\min} N_0 / (2P_s)$.

Conversion for neural LFP signals: Measure background noise variance σ_n^2 (V^2) over bandwidth B (Hz) from spontaneous recordings. Then $N_0 = \sigma_n^2 / B$. For a local field potential oscillation with RMS amplitude A_{LFP} (V) in matched band, signal power is $P_s \approx A_{\text{LFP}}^2 / (2R)$ where R is effective electrode impedance. Typical values: $\sigma_n^2 \sim 10^{-10} \text{ V}^2$, $B \sim 1000 \text{ Hz}$, $N_0 \sim 10^{-13} \text{ W/Hz}$; $A_{\text{LFP}} \sim 10 \mu\text{V}$, $R \sim 10^5 \Omega$, $P_s \sim 10^{-12} \text{ W}$. For $\rho_{\min} = 5$, $\tau_{\text{SNR}} \sim 0.25 \text{ ms}$.

ESM 2: Derivation of dimensional chemical potential

Let $E = E(S, V, N, \mathcal{D})$ be the fundamental thermodynamic relation with realized dimensional structure $\mathcal{D}$ as a state variable. The extended first law is

$$\text{d}E = T \text{d}S - p \text{d}V + \mu \text{d}N - \lambda_{\mathcal{D}} \text{d}\mathcal{D}$$

defining dimensional chemical potential

$$\lambda_{\mathcal{D}} = - \left(\frac{\partial E}{\partial \mathcal{D}} \right)_{S, V, N}$$

Passing to entropy representation $S = S(E, V, N, \mathcal{D})$:

$$\lambda_{\mathcal{D}} = T \left(\frac{\partial S}{\partial \mathcal{D}} \right)_{E, V, N}$$

For metric ε -entropy $S_\varepsilon = k_B \ln N_\varepsilon$ (Minkowski covering number), empirically $\partial S_\varepsilon / \partial \mathcal{D} \approx k_B s_\varepsilon$ with $s_\varepsilon \sim \mathcal{O}(1)$. Thus $\lambda_{\mathcal{D}} \approx k_B T s_\varepsilon$. For equilibrium with $s_\varepsilon \approx 1$: $\lambda_{\mathcal{D}} \sim k_B T$. For non-equilibrium: $\lambda_{\mathcal{D}} \sim k_B T_{\text{eff}}$ where T_{eff} is effective temperature from fluctuation-dissipation violations.

ESM 3: Proposed Kuramoto network validation protocol

To empirically validate Eq. 4, we propose the following simulation protocol for future work:

Setup: Simulate networks of $N = 500$ – 2000 coupled phase oscillators:

$$\frac{d\theta_i}{dt} = \omega_i + \frac{K}{N} \sum_{j=1}^N \sin(\theta_j - \theta_i)$$

with natural frequencies ω_i drawn from uniform distribution (width $\Delta\omega$), randomized initial phases, and fourth-order Runge-Kutta integration (timestep $dt = 0.01/\Delta\omega$).

Measurement: For coupling near critical value $K \approx K_c + 0.1K_c$, measure time between transient alignments (order parameter $r(t)$ exceeding threshold r_* for duration $> \varepsilon/\omega_0$). Test whether alignment time scales as $\langle \tau_{\text{align}} \rangle \propto (\varepsilon/2\pi)^{-\alpha(1-r)(M-1)}$ where M is the number of weakly-coupled clusters (via community detection on phase-coupling adjacency matrix).

Predictions:

- Topology-dependent α : all-to-all ~ 0.9 , sparse random (Erdős-Rényi, $\langle k \rangle = 20$) ~ 0.7 , modular (Girvan-Newman) ~ 0.6
- Circular variance $(1 - r)$ should provide better fit than $(1 - r^2)$ at moderate coherence ($r \approx 0.5$ – 0.7)
- Scaling exponent $(M - 1)$ should match number of semi-independent modules from community structure

Code implementing this protocol will be made available upon publication.

ESM 4: Figure legends

Figure 1. Hourglass architecture. High-dimensional continuous dynamics (top) funnel through four physical bottlenecks (middle) to produce low-dimensional commits (bottom). The slowest constraint (max operation) sets observable temporal grain. Quantum and SNR limits typically negligible for neural systems; coherence time dominates.

Figure 2. Speed-flexibility frontier. Commit rate (Hz) versus coordination depth M for different coherence values r . Higher r enables faster processing but restricts exploration to low-dimensional synchronized manifolds. Biological examples overlaid: mind-wandering (high- M /low- r), focused attention (moderate), automaticity (low- M /high- r). Arrows show typical trajectories from training and arousal.

Figure 3. Parameter sensitivity heatmap. $\text{Log}_{10}(\tau_{\text{coh}})$ versus Kuramoto coherence r (x-axis) and coordination depth M (y-axis) at fixed $\varepsilon = \pi$, $\Delta\omega = 2\pi \times 10$ rad/s, $\alpha = 0.9$. Color scale spans -3 to $+3$ (milliseconds to kiloseconds). Contours show orders-of-magnitude changes from modest parameter shifts, explaining biological dynamic range without proportional metabolic costs.